

Stefano Racioppi, Ph.D.

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Appointments and Professional Experience

09/2025 – present Research Associate, *University of Cambridge*, UK.
Group: Prof. Chris Pickard.
03/2021 – 08/2025 Postdoctoral researcher, *State University of New York at Buffalo*, USA.
Group: Prof. Eva Zurek.
07/2023 – 09/2023 Visitor, *Aalto University*, Finland.
Group: Prof. Patrick Rinke.
02/2019 – 02/2021 Postdoctoral researcher, *Chalmers University of Technology*, Sweden.
Group: Prof. Martin Rahm.



Education

10/2015 – 01/2019 Ph.D. in Industrial Chemistry, *University of Milan*, Italy.
10/2016 – 03/2018 Visiting student at the *University of Bern*, Switzerland.
10/2013 – 10/2015 Master's degree in industrial chemistry, *University of Padua*, Italy.
10/2010 – 09/2013 Bachelor's degree in industrial chemistry, *University of Padua*, Italy.

Research Interests

Computational and Theoretical Chemistry – High Pressure Chemistry – Crystal Structure Prediction – Quantum Crystallography – Chemical Bonding – Materials and Planetary Science – Superconductivity – Machine-Learning.

Distinctions

07/2026 Associate Fellow of the Higher Education Academy.
02/2026 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, Seal of Excellence 2026.
01/2026 Research Development Fellows Program – University of Cambridge.
03/2025 Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, Seal of Excellence 2025.
09/2019 Best Ph.D. Thesis Award by the Italian Crystallographic Association (AIC).
07/2017 Outstanding poster prize offered by Wiley – Journal of Computational Chemistry and International Journal of Quantum Chemistry. Workshop on ab Initio Modeling in Solid State Chemistry with CRYSTAL, *Minneapolis*, USA.

Fundings, Proposals and Scholarships

05/2025 **Awarded:** 1 shot-day (8 shots in total) at Omega Laser Facility at the Laboratory for Laser Energetics (LLE). Project: *Structural Determination of Al₂O₃ at super-Earth Conditions – AIO Discovery*. PI: Dr. Racioppi S.
04/2025 **Awarded:** Beamtime (18 shifts) at the European Synchrotron Radiation Facility (ESRF). Project: *Diagnosing Electrified Formation in Ca up to 40 GPa with XANES and SCXRD*. PI: Dr. Chin, D. A.; co-PI: Dr. Racioppi, S.
04/2024 **Awarded:** 34th European Crystallographic Meeting Scholarship; € 300.
09/2023 **Contributed:** *High Energy Density Quantum Matter - Renewal*; Award number: LOI-0000043230; Source of Support: Department of Energy (DOE); PI: Dr. Collins G.W.; Co-PIs: Dr. Deemyad S.; Dr. Hemley R. J.; Dr. Zurek E.; \$ 2,100,000.
02/2023 **Awarded:** Beamtime at Laser-lab Europe LENS; *Tuning Electrical Conductivity of Aurophilic Interactions by Compression* (ID: 22396); PIs: Dr. Racioppi S., Dr. Poręba T.; Prof. Macchi P.
09/2022 **Awarded:** *Seed Funding Project* – Center for Matter at Atomic Pressure (CMAP); \$ 3500.
10/2018 **Awarded:** Scholarship; National Interuniversity Consortium of Materials Science and Technology (INSTM); PI: Prof. Sironi A.; € 5862.
05/2018 **Awarded:** Travelling Grant; Ettore Majorana Foundation; € 600.

Publications

H-index: 10 | Total number of citations: 335

- (31) Liu, Z.; **Racioppi, S.**; Hilleke, K. P.; Pandit, A.; Ma, S.; Hermann, A.; Yan, D.; Zurek, E.; Miao, M. Non-local Chemistry Driven by Cation-Anion Size Disparity in Helium Inserted Compounds under High Pressure, *arXiv:2511.23400*.
- (30) Farraj, H.; **Racioppi, S.**; Garbarino, G.; Ahart, M.; Mondal, A.; Parra, S. G.; Smith, J. S.; Cohen, R. E.; Zurek, E.; Cabana, J.; Hemley, R. Novel Pressure-Induced Transformations of PbTiO₃, *arXiv:2511.05756*.
- (29) **Racioppi, S.***; Zurek, E. Quantum Effects or Theoretical Artifacts? A Computational Reanalysis of Hydrogen at High-Pressure, *arXiv:2510.02098*.
- (28) Ocampo, I. K.; Kim, D.; Smith, R. F.; Coppari, F.; **Racioppi, S.**; Zurek, E.; Rygg, J. R.; Chin, D. A.; Millot, M.; Eggert, J. H.; Duffy, T. S.; Atomic-level structure and pressure-density response of Fe-O compounds at deep Earth and exoplanetary interior conditions, *under review in Physics of the Earth and Planetary Interiors*.
- (27) **Racioppi, S.**; Kelly, A.; Zurek, E. Computational Investigation of Al₂O₃ Structures at Super-Earth Interior Conditions, *ACS Earth Space Chem.*, **2026**.
- (26) Weadock, N.; Holle, W.; Myford, K.; **Racioppi, S.**; Ghimire, A.; Ladd, D.; Park, C.; Zurek, E.; Toney, M.; Bridges, F.; Deemyad, S. Pressure-driven helium insertion for structural stability of CH₃NH₃PbBr₃ hybrid perovskites, *Chem. Mater.*, **2026**.
- (25) Caussé, M.; Bozier, K.; Cooke, P. I. C., **Racioppi, S.**; Pickard C. J. Metastability and high-T_c superconductivity in Al₅-type ternary hydride YSbH₆ at moderate pressure, *PRB*, **2026**.
- (24) Coleman, A. L.; Singh, S.; Lockard, T.; Ocampo, I. K.; Lazicki, A. E.; Gorman, M. G.; **Racioppi, S.**; Krygier, A. G.; Wehrenberg, C. E.; Ahmad, R.; Hamel, S.; Han, S.; Ginnane, M. K.; Swift, D. C.; Bonev, S. A.; Zurek, E.; Duffy, T. S.; Eggert, J. H.; McNaney, J.; Smith, R. F. Body-centered cubic phase transformation in gold at TPa pressures. *PRL*, **2025**.
- (23) **Racioppi, S.***; Zurek, E. Using Topology to Predict Electrides in the Solid, *J. Phys. Chem. A*, **2025**.
- (22) **Racioppi, S.***; Zurek, E. Activation of Semicore Electrons in Alkali Metals and Their Role in the B1–B2 Phase Transition under Pressure, *J. Am. Chem. Soc.*, **2025**, *147*, 36, 32745-32751.
- (21) Storm, C. V.; **Racioppi, S.**; Duff, M. J.; McHardy, J. D.; Zurek E.; McMahon, M. I. Experimental Evidence of Interstitial Electron Density in Transparent Dense Sodium, *Commun. Mater.*, **2025**, *6*, 201.
- (20) **Racioppi, S.**; Zurek, E. High-Pressure Electrides: A Quantum Chemical Perspective, *Annu. Rev. Mater. Res.*, **2025**, *55*, 421-442.
- (19) **Racioppi, S.**; Saffarin-Deemyad, I.; Holle, W.; Belli, F.; Ferry, R.; Kenney-Benson, C.; Smith, C. J.; Zurek, E.; Deemyad, S. Phase Lithium's low-temperature phase transitions: Insights into quantum lattice dynamics and superconductivity, *PRB*, **2025**, *111*, 054111.
- (18) **Racioppi, S.**; Zurek, E. Looking at High-Pressure Electrides Through the Lens of Quantum Crystallography: The Case of Simple Cubic Calcium, *Acta Cryst.*, **2025**, B81.
- (17) **Racioppi, S.**; De la Roza, A.; Hajinazar, S.; Zurek, E. Powder X-Ray Diffraction Assisted Evolutionary Algorithm for Crystal Structure Prediction, *Digit. Discov.*, **2025**, *4*, 73-83.
- (16) **Racioppi, S.**; De la Roza, A.; Hajinazar, S.; Zurek, E. PXRD-Assisted Crystal Structure Predictions, *Acta Cryst.*, **2024**, A80, e396.
- (15) **Racioppi, S.**; P.; Hyldgaard, P.; Rahm, M. Quantifying Atomic Volume, Partial Charge and Electronegativity in Condensed Phases, *J. Phys. Chem. C*, **2024**, *128*, 4009-4017.
- (14) **Racioppi, S.***; Storm, C. V.; McMahon, M. I.; Zurek, E. On the Electride Nature of Na-hP4, *Angew. Chem. Int. Ed.* **2023**, e202310802.
- (13) **Racioppi, S.**; Miao, M.; Zurek, E. Intercalating Helium into A-site Vacant Perovskites, *Chem. Mater.* **2023**, *35*, 4297-4310.
- (12) Fang, M.; Srikanth Kumar, G.; **Racioppi, S.**; Zhang, H.; Zurek, E.; Lin, Q. Hydrazonyl Sultones as Stable Tautomers of Highly Reactive Nitrile Imines for Fast Bioorthogonal Ligation Reaction, *J. Am. Chem. Soc.* **2023**, *145*, 18, 9959-9964.
- (11) **Racioppi, S.**; Lolur, P.; Hyldgaard, P.; Rahm, M. A Density Functional Theory for the Average Electron Energy, *J. Chem. Theory Comput.* **2023**, *19*, 799-807.
- (10) Antle, J.; Kimura, M.; **Racioppi, S.**; Lang, M.; Damon, C.; Gatley-Montross, C.; Sánchez B., L.; Miller, D.; Zurek, E.; Brown, A.; Gast, K.; Simpson, S. Applying Density Functional Theory to Common Organic Mechanisms: A Computational Exercise, *J. Chem. Educ.* **2023**, *100*, 355-360.
- (9) **Racioppi, S.**; Orian, L.; Gennaro, A.; Isse, A. Solvent Coordination Effect on Copper-Based Molecular Catalysts for Controlled Radical Polymerization, *Catalysts* **2022**, *12*, 1656.

- (8) Poręba, T.; **Racioppi, S.***; Garbarino, G.; Morgenroth, W.; Mezouar, M. Investigating the Structural Symmetrization of CsI₃ at High Pressures through Combined X-ray Diffraction Experiments and Theoretical Analysis, *Inorg. Chem.* **2022**, 61, 10977-10985.
- (7) Della Pergola, R.; Garlaschelli, L.; Macchi, P.; Ruffo, R.; **Racioppi, S.**; Sironi, A. From Small Metal Clusters to Molecular Nanoarchitectures with Core-Shell Structure: Synthesis, Redox Fingerprint, Theoretical Analysis and Solid-State Structure of [Co₃₈As₁₂(CO)₅₀]⁴⁺, *Inorg. Chem.* **2022**, 61, 9888-9896.
- (6) Kumar, G. S.; **Racioppi, S.**; Zurek, E.; Lin, Q. Superfast Tetrazole–BCN Cycloaddition Reaction for Bioorthogonal Protein Labeling on Live Cells, *J. Am. Chem. Soc.* **2022**, 144, 57–62.
- (5) **Racioppi, S.**; Rahm M. In-Situ Electronegativity and the Bridging of Chemical Bonding Concepts, *Chem. Eur. J.* **2021**, 27, 18156-18167.
- (4) **Racioppi, S.***; Sironi, A.; Macchi, P. On Generalized Partition Methods for Interaction Energies, *Phys. Chem. Chem. Phys.* **2020**, 24291-24298.
- (3) **Racioppi, S.**; Andrzejewski, M.; Colombo, V.; Sironi, A.; Macchi, P. Different Metallophilic Attitudes Revealed by Compression, *Inorg. Chem.* **2020**, 59, 2223–2227.
- (2) **Racioppi, S.***; Della Pergola, R.; Colombo V.; Sironi, A.; Macchi, P. Electron Density Analysis of Metal Clusters with Semi-Interstitial Main Group Atoms. Chemical Bonding in [Co₆X(CO)₁₆][−] Species, *J. Phys. Chem. A* **2018**, 122, 5004-5015.
- (1) Della Pergola, R.; Sironi, A.; Colombo, V.; Garlaschelli, L.; **Racioppi, S.**; Sironi, A.; Macchi, P. Periodical trends in [Co₆E(CO)₁₆][−] clusters: Structural, synthetic and energy changes produced by substitution of P with As, *J. Organomet. Chem.* **2017**, 849-850, 130-136.

* Corresponding author

Talks

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| 04/2027 | Invited. American Physical Society (APS) Global Physics Summit; Computational Physics. |
| 08/2026 | Invited. 27 th Congress and General Assembly of the International Union of Crystallography; Microsymposium MS084: Quantum crystallography of non-equilibrium or nanostructured materials. |
| 06/2025 | Contributed. 24 th Biennial International Conference on the Science of Compression in Condensed Matter (SCCM): Computational Methods: Theory, Codes and Applications (FMDA/B). |
| 06/2025 | Contributed. Crystal Structure Prediction Workshop, a Practical Approach, Poitiers. |
| 09/2024 | Contributed. 4 th European Symposium on Chemical Bonding (CBOND2024). |
| 08/2024 | Contributed. 34 th European Crystallographic Meeting (ECM34); Microsymposium M27: High Pressure Crystallography: Exploring Structure and Method Development at Extreme Conditions. |
| 07/2024 | Invited. Gordon Research Seminar (GRS), Research at High Pressure. |
| 07/2023 | Contributed. 28 th AIRAPT and 60 th EHPRG International Conference on High Pressure Science and Technology; Session: Chemical Bonding. |
| 03/2022 | Contributed. American Physical Society (APS) March Meeting; Session G24: Matter at Extreme Conditions: Phase Transitions. |
| 09/2021 | Invited. Italian Crystallographic Association (AIC) XLIX Meeting. |
| 06/2018 | Contributed. International School of Crystallography in Erice; 52 nd Course: Quantum Crystallography. |

Seminars

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| 10/2026 | <i>Centre for Science at Extreme Conditions (CSEC) Seminar</i> , University of Edinburgh, <i>Host</i> : Dr. Dominique Laniel. |
| 04/2026 | <i>Diamond's Science Happy Hour</i> , Diamond Light Source Ltd; <i>Host</i> : Dr. Annette Kleppe. |
| 03/2026 | Italian Young Crystallographers (GCI) <i>GCI Insights</i> Virtual Seminar; <i>Host</i> : Dr. Martina Lippi. |
| 11/2025 | Chicago/DOE Alliance Center (CDA) Virtual Seminar Series; <i>Host</i> : Prof. Russell Hemley. |
| 05/2025 | Cambridge Crystallographic Data Centre (CCDC), Virtual Seminar; <i>Host</i> : Dr. Fabio Montisci. |
| 07/2024 | High Energy Density Science Center (HEDS) Virtual Seminar Series; Lawrence Livermore National Laboratory (LLNL, USA); <i>Host</i> : Dr. Federica Coppari. |
| 01/2024 | University of Milan (Italy), Dep. of Chemistry; <i>Host</i> : Prof. Davide Proserpio. |
| 08/2023 | Aalto University (Finland), Dep. of Physics; <i>Host</i> : Prof. Patrick Rinke. |
| 06/2023 | University of Oviedo (Spain), Dep. of Chemistry; <i>Host</i> : Prof. José Manuel Recio. |
| 04/2023 | Italian Young Crystallographers (GCI) <i>Happy Hour</i> Virtual Seminar; <i>Host</i> : Dr. Marta Morana. |
| 03/2022 | Chicago/DOE Alliance Center (CDA) Virtual Seminar Series; <i>Host</i> : Prof. Russell Hemley. |

Services

- 07/2026 Chair of the Gordon Research Seminar (GRS), Research at High Pressure.
- 07/2026 Co-organizer and chair of the “*Under Pressure: from Chemical Bonding to Planetary Structures and Evolution*” session at the Goldschmidt 2026 conference.
- 03/2026 Appointed Springer Nature Reviewing Editor of Nature Portfolio journals.
- 07/2025 Co-organizer of the “*High Energy Density Physics: Quantum Materials and Quantum Phenomena*” session at the 24th Biennial Conference on Science of Compression in Condensed Matter (SCCM).
- 06/2025 Co-organizer of the workshop on the crystal structure prediction code XtalOpt at the Crystal Structure Prediction Workshop, a Practical Approach, held in Poitiers, France.
- 06/2022 Co-organizer of the workshop on the crystal structure prediction code XtalOpt at the International School of Crystallography in Erice; 56th Course: Crystallography under extreme conditions.
- 06/2022 Co-organizer of the workshop on computational chemistry techniques at the International School of Crystallography in Erice; 56th Course: Crystallography under extreme conditions.

Teaching

- 2025 – 2026 Demonstrator for a course in the Scientific Computing MPhil on Electronic Structure and Density Functional Theory – University of Cambridge.
- 2024 Special virtual lecture for the general chemistry course at the University of Milan – “*History and Evolution of the Electronegativity*”
- 2022 – 2023 Online workshops for the usage of the crystal structure prediction code XtalOpt.
- 2018 – 2019 Teaching assistant in physical chemistry II for 2nd year degree in Industrial Chemistry – University of Milan.
- 2015 – 2016 Teaching assistant in general and inorganic chemistry for 3rd year degree in Physics – University of Milan.

Supervision

- 2026 Supervisor of one student for the MPhil project in Scientific Computing (Joch Cheung) and of two students for the MPhil written assignment in Scientific Computing (Rongze Sun, Jiawei Li) – University of Cambridge.
- 2021 – 2025 Supervisor of two undergraduate students (Amelia Kelly, Jarrod DuMond) – SUNY Buffalo.
Co-supervisor of three PhD students (Morgan Reddington, Masashi Kimura, Isaac Ampersand-Kwadwo Akwetey) – SUNY Buffalo.
- 2021 Supervisor of a high school student (Henry Feng) for summer project – SUNY Buffalo.
- 2017 – 2019 Co-supervisor of two master students (Stephanie Terruzzi and Simona Sorbara) – University of Milan.

Memberships

- 2026 – present European Committee for Chemical Bonding.
- 2025 – present Trinity Hall College of Cambridge.
- 2024 – present European Crystallographic Association (ECA).
- 2021 – present American Physical Society (APS).
- 2018 – 2026 Italian Crystallographic Association (AIC).

Computational Developments and Codes

Electronegativity in gas-phase: <https://github.com/SteRacioppi/Chi-bar-Atomic-and-Molecular-Electronegativity>

Electronegativity in solid: https://www.scm.com/doc/BAND/Expert_Options/Restarts.html

Public Service and Outreach

- 2026 *Knowable Magazine*, [Meet the mysterious electrified](#).
- 2025 *News Center - News & information from UB*, [Core electron bonding may not always require extreme pressure, study finds](#).
- 2025 *Science News*, [A new iron compound hints ‘primordial’ helium hides in Earth’s core](#).
- 2023 *News Center - News & information from UB*, [Sodium’s high-pressure transformation can tell us about the interiors of stars, planets](#).